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Wasana Bandara & Rehan Syed

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The role of a protocol in a systematic literature review

Wasana Bandara and Rehan Syed

School of Information Systems, Queensland University of Technology, Queensland, Australia

ABSTRACT

A good literature review that effectively synthesises current knowledge can contribute to the cumulative knowledge growth and advancement of a field. This tutorial-paper complements the existing methodological guidelines, with a particular focus on the 'literature review protocol'- which can significantly improve the quality, efficiency, and impact of a literature review endeavour. In this paper, we describe what a 'literature review protocol' is, discuss its major components, and provide essential design guidelines. In particular, we provide a comprehensive toolkit with step-by-step instructions, templates, and illustrative examples that users may adapt to their own projects. This information is appropriate for novice and expert researchers conducting a literature review for publication purposes, and also for reviewers or editors who assess literature review papers. It is also beneficial for educators engaged in research trainings.

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1. Introduction

Literature reviews (when executed correctly) are increasingly recognised as valuable contributions to knowledge. A good literature review that effectively synthesises the current knowledge can contribute towards the cumulative knowledge growth and progression of a field (Li & Chan, 2019; Paré et al., 2016; Webster & Watson, 2002). As a result, many journals and conferences have become far more receptive to literature review papers, with some even having genres/departments/tracks dedicated solely to literature reviews. With the demand for literature reviews increasing, there has been a rise in published literature review methodological guidelines aiming to provide researchers the required guidance on how to conduct a literature review effectively. They each cover diverse aspects of a literature review journey. In this tutorial-paper, we add to this methodological know-how, with a dedicated discussion on the **role of a protocol** within a literature review effort¹.

We introduce the notion of a literature review protocol as a critical tool that all researchers engaged in a systematic literature review should develop and use. We position the literature review protocol as a detailed blueprint of the end-to-end literature review journey. The protocol is a valuable planning and quality assurance tool which can provide more structure to the literature review process and can be

CONTACT Wasana Bandara  w.bandara@qut.edu.au  School of Information Systems, Queensland University of technology, Queensland, Australia

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used as a ‘living’ document throughout the lifetime of the literature review. A literature review protocol can also serve as a collaboration tool when working with a team of other researchers. We describe diverse benefits (see [Section 2](#)) and present detailed guidelines (see [Section 3](#)) outlining the key elements that should be included in a literature review protocol. We complement our guidelines with a literature review protocol template ([Appendix A](#)), and ancillary material (see [Appendix B](#)) consisting of additional resources that aim to assist our readers to rapidly progress in their literature review protocol design and execution².

Our paper targets researchers interested in conducting and publishing literature reviews in the field of Information Systems (IS). Hence, it draws upon and refers to examples from the IS domain. Nevertheless, these guidelines can also be adapted to other fields. The content presented in this paper is most appropriate for novice researchers conducting a literature review for publication purposes, but it may also serve as a useful reference for experienced researchers as well. Reviewers of literature review papers can also benefit from this paper, as it serves as a checklist and a useful reference when providing developmental feedback to the authors. This tutorial-paper will also benefit educators engaged in research trainings, as its detailed guidelines, extended examples, and ancillary material may be effectively used as training resources.

The content presented in this paper provides practical suggestions that are based in our experiences and knowledge of literature reviews, of which we have published over twenty in reputed outlets. Additionally, in order to enrichen our own guidelines, we build on prior literature review methodology guidelines, and extracted further knowledge from other papers about protocols in literature reviews (which are quite rare).

We would like to make several disclaimers before we proceed. First, we are using our own work here as examples. Our intention is not to position this work as superior to other published literature review – rather, we hope that our knowledge on literature reviews, which includes protocols surrounding design, evolution etc, can be used effectively so as to provide appropriate illustrative case examples in this paper. Second, this paper caters to those who have had at least a minimal amount of exposure to systematic literature reviews and have some familiarity with the overall process. Given that the proposed literature review protocol is positioned as a tool that supports the end-to-end literature review journey, having familiarity with this journey is a necessity. We provide several highly recommended literature review methodological papers in Part A of [Appendix B](#), in order for what we present here to be more valuable to readers who are at the beginning of their literature review journey. We acknowledge the many other well-established literature review guidelines that exist, and mean no bias in selecting the few that we recommend here. They were chosen simply based on best fit-for-purpose, and our choices were influenced by our own familiarity of these guidelines through using them in our own prior work.

The remainder of the paper first presents some contextual background, including highlighting the motivation of this tutorial-paper. It then delves into outlining the key elements of a literature review protocol. The paper concludes with a summary and some further recommendations. A separate ‘Ancillary Material’ file is made available (as part of [Appendix B](#)), which contains information that supplements the main paper (see Literature Review Protocol – Google Drive).

2. Background and motivation

A literature review protocol is a predetermined, detailed plan that describes how the review will be conducted. The protocol is akin to a detailed ‘road map’ of a research study design. Protocols are developed in the planning phase of a systematic literature review, and once developed, become ‘living documents’ that will constantly be updated throughout the literature review conduct to help the researchers capture and report on the review process.

We argue that a literature review protocol should cover the end-to-end journey of the entire literature review effort. This includes the review’s context and purpose, the questions intended to be answered by the literature review, a detailed literature search strategy that articulates how the papers included in the review are selected, an explanation on how the papers are to be coded and analysed, and some indication as to how the final findings are to be presented. Other aspects, such as the overall quality assurance efforts to be deployed throughout, should also be articulated.

The review protocol is critically essential for extensive systematic reviews (Brereton et al., 2007; Okoli & Schabram, 2010). It is required for improving the quality of a review since it decreases the likelihood of researcher bias in data selection and analysis (Kitchenham et al., 2009). It also boosts the review’s credibility because others can repeat the study using the same technique for cross-checking and verification. Articulating the detailed design in a protocol (as recommended herein) enables researchers to think carefully and deeply about upcoming steps, potential challenges, and pre-plan a quality-assured design accordingly.

A protocol helps researchers working in teams to maintain a straightforward and structured process so as to ensure that all procedures and activities are understood by the team players (DeCuir-Gunby et al., 2011). It also assists research teams with increasing their efficiency, clarity, and transparency, and helps with team effort in multi-researcher projects (Leidner, 2018; Syed et al., 2020). A protocol can help to bolster the confidence and efficiency of the research team, especially if there are novices in literature reviews involved.

Despite the importance of a protocol and communicating its design and operationalisation, only a handful of existing published research even mention a protocol, let alone share the details of the protocol used. Bharosa et al. (2009); Li and Chan (2019); Syed et al. (2018); Syed et al. (2020) are some examples of published literature reviews in Information Systems outlets that report on the protocol being used. In these papers, it can be clearly seen how they present the protocol development and communication as an integral component within the overall study design (e.g. Figure 2 of Li & Chan, 2019, p. 9; and Figure 1 of Syed et al., 2018, p. 8). While these form good illustrative examples, they do not provide any detailed guidance on the design and use of a protocol, a gap that this paper aims to address.

Furthermore, existing discussions of the protocols within literature reviews only cover the coding phase. We argue that the protocol should cover a larger scope than just coding; it should cover the end-to-end process of the entire literature review effort (i.e. setting the goals, looking for the *right* papers, coding, interpretations and presentations of the findings). This is important because the coding is influenced by the input that goes in (i.e. the paper selection) and by how one wishes to present the findings. Thus, it makes

sense to have a single protocol that encompasses all these stages in a holistic manner. This will allow all stages of the review process to benefit from having a protocol.

The guidelines in this paper are predominantly applicable to works in which published literature serves as the primary source of data for the literature review. But the learnings from here can also be applied to multi-phase studies, where literature reviews form one (or more) of the core phases of a larger multi-phased study. For example, Bharosa et al. (2009) report on how a protocol was used across a three-phased approach, which commenced from a literature review and continued with case surveys and interviews.

We acknowledge that there are diverse literature review types. Paré et al. (2015) provide a detailed synthesis on literature review types in IS and how these different types call for diverse styles and attention to different elements. Booth et al. (2016) and Cooper (1988) are other popular literature review typologies. While the guidelines we present are generic and adaptable to all review types, we encourage you to determine the type of literature review for your work and communicate that to the readership early on. Not only does this help to set the right expectations for your readers on what aspects to emphasise when communicating the literature review outcomes, but it also gives you the appropriate context on which to adjust the proposed generic protocol according to the relevant literature review type. To illustrate further, taking Paré et al. (2015) literature review types as an example, the protocol of a 'Scoping review' will need to pay much attention to clarifying and justifying the breadth of literature covered, and that of a 'Theoretical review' will need to more precisely define how themes were captured, coded and interpreted in the coding process, and be accurately reported. On the other hand, a 'Narrative' review may not need to report on the paper searching and screening at all, but having such documentation in the protocol is still useful to the researchers for completeness and quality checking

Appendix A provides a template of a literature review protocol that can be adapted and used by our readers. Appendix B, part B, presents an illustrative case example where we introduce a published literature review study and discuss its protocol, detailing how it supported the resulting study outcomes.

3. Key elements of a literature review protocol

We summarise the essential elements of a literature review protocol into several parts, namely, (i) goal definition, (ii) topic area(s) of interest, (iii) search strategy, (iv) paper screening, (v) tool use, (vi) coding plans (including coding rules and code books), and (vi) reporting plans. Each element is outlined below.

3.1. Goal(s) of the literature review

It is vital to establish early on what the goals of your literature review are. This 'sets the scene', which helps with determining the scope and boundary conditions and can influence many of the proceeding steps of a literature review. For example, the purpose of the literature review could influence the decisions on where to search for your papers. A study that aims to understand the current status of constraint-based innovation within the IT sector will have a different scope to a study that aims to define what constraint-based innovation is. The first can already be narrowed down specifically for

papers in relation to IT (hence conduct its search in IT outlets), whereas the second needs to ideally capture a broader scope of papers that discuss constrained-based innovation more widely.

The goal(s) should be articulated in the protocol in the form of simple statements or research questions (see point # 1 in [Appendix A](#)). In addition to listing research questions or goal statements, we recommend that the researcher takes some time to write a few lines that describe their intentions. This is because articulating the broader intentions could shed light (at present or later) onto what other questions the literature review might help answer.

3.2. Topic areas or key phenomena of interest

Once the goals of your literature review have been outlined, the next step is to identify the key phenomena that you are interested in exploring within the review. Recognising the key phenomena of interest contributes towards the scoping of your literature review and is also important for the steps ahead as this action forms the first step towards identifying your keywords for your search strategy. It will also contribute towards setting up inclusion/exclusion rules later down the track.

Some literature reviews may have multiple phenomena that are being investigated together; others, on the other hand, may have only one. For example, if a study is about how *'Information Systems can support (or inhibit) constraint-based innovation'*, then there are two key phenomena of interest (a.k.a. key topic areas): (i) constraint-based-innovation and (ii) information systems. In this case, the authors would be interested in studies that discuss *both* of these phenomena.

Once the key topic areas are determined, we then recommend that they are tabulated and synonyms or similar terms that may capture related concepts are recognised (see #2 of [Appendix A](#)). This becomes an important input towards the design of the search terms.

The refinement of the synonyms and/or related concepts is an iterative process. Whilst it is good to try to have a solid list at the start, it is also important to be able to adjust these. For example, once you start searching, you may identify that there are new synonyms or new related concepts that you didn't consider early on, in which case they should be added in – this process should be repeated accordingly.

We recommend that the key topics (including synonyms and related concepts) are discussed and confirmed by a broader team beyond a single researcher. We consider this is to be an essential point for corroboration, where, ideally, a subject matter expert in the chosen topic areas is consulted.

3.3. The search strategy

Deriving the search strategy is an iterative process. It is an important prerequisite to determine some of the boundary criteria, such as where to conduct your search in. But it is also important to be mindful that this will evolve and refine the protocol as one goes through multiple rounds of executing the search.

Key concepts that must be thought about and communicated about the search strategy are:

- Where do you conduct your search in? Often people search within databases, but one can also search in diverse outlets depending on the scope of the literature review. For example, in Felt et al. (2014), which looked at the status of shared services literature specifically within the Information Systems discipline, the authors chose several prominent Information Systems journals and conferences in which they would do their search. On the other hand, in the example of Syed et al. (2020) in their inquiry about the status of Robotic Process Automation (RPA), they looked at a broad set of databases that were likely to cover papers on process automation or automation in general.

Sometimes you might combine both databases and selected outlets. A key aspect here is to be able to clearly justify your decisions. This articulation should cover the alignment and completeness. For example, the alignment of the topic area to the disciplinary coverage of the selected databases, and how the combination of the targeted databases should cover most (if not all) anticipated papers on the topic of interest. A good systematic literature review should articulate precisely how they choose the sources to conduct their search and the related rationale behind these decisions in the research design section.

While we predominantly covered the *sources* in which the searches are to occur, this also should cover the *fields* that you choose the search to be done in. These should be listed and justified (see #3.1 of [Appendix A](#)). Often searches are done in title, keywords, and abstract fields, but depending on the scope and the nature of the review, you could choose only one or two of the above, or perhaps expand the scope to do a full-body search in the papers. Whichever options you choose, they need to be captured and justified. Aspects such as feasibility will play a role here as well and are accepted facts to confine a search, but need to be articulated.

- Criteria for initial search: Every search is bounded by some criteria. Common criteria include things like the papers being peer-reviewed, having access to the full paper, the language (e.g. only papers in English). Sometimes you might extend these basic criteria to help with further scoping the work. For example, you might say that you will only extract papers after a certain time (e.g. papers published after the year 2000). It is very important to list these criteria and to provide justification for these choices (see #3.2 of [Appendix A](#)).

Note that criteria setting happens at multiple stages of the literature review. It can occur when you are initially looking at designing your search strategy, and also later, when you have a set of papers you have already extracted and you are seeking to determine inclusion/exclusion based on degrees of relevance (see [Section 3.4](#) below). We like to treat these stages differently; criteria setting at the initial search stages is very much aligned with the search strings that we create. The inclusion/exclusion criteria constitute a later step which occurs once you have extracted the papers and are determining if the papers are to be included in the pool of papers to be analysed.

- Search strings: You will design search strings that will use the key topics, synonyms and related concepts identified above, applying typical Boolean logic such as 'AND'/'OR' functions. It is important to familiarise yourself with each of the outlets and how their interfaces are set, for the search strings can differ accordingly. For example, the syntax used for wild cards in one database can be different to another. Hence, you should not assume that the same search string can be used in all databases – they will need to be adjusted.

We recommend that you have a look at the advanced search facilities for the databases you use and set up the search strings accordingly. We also strongly recommend that you discuss the derived search strategy with a reference librarian

(which most institutions provide access to for researchers).

The progressive results of you applying the search strings across the different search outlets should be logged and then reflected upon (see 3.3 of [Appendix A](#)). The search results at this stage could lend further insights into any adjustments in the topic areas searched for and overall changes that should be considered for the search string. At times this means refining and repeating the above-mentioned steps.

3.4. Paper screening

The steps up to now are designed with a focus on maintaining completeness, which means we try as hard as possible to gather *all* the relevant papers. Hence, a much broader scope is maintained. However, after the set of papers are extracted, they need to be screened.

Paper screening often happens in multiple stages. For example, as you extract the papers, you might decide to deeply review the abstracts and make some inclusion/exclusion decisions early on. Or, after this or instead of this step, one may consider screening the papers at the same time they start the coding process. Thinking upfront about how the paper screenings will occur and setting the operational processes in place is deeply recommended. For each point of screening, it is important to note and articulate the heuristics (or the rules) on which you will decide to exclude or include papers (see point # 4 of [Appendix A](#)).

Once again, this step is another place where corroboration with other researchers is recommended. You would ideally discuss and agree upon the inclusion/exclusion criteria and clarify where in the process the assessment takes place. It is recommended (e.g. Kitchenham et al., 2009) to take a sample of the literature (ideally approx.20–30 papers) and engage all coders simultaneously to determine which articles should be included or excluded. The outcomes from here can also be used to re-specify or fine-tune the relevance and/or quality criteria used to screen the papers.

When approaching full set of papers, it is very likely that you will find papers which are difficult to assess (especially in the early stages, such as when you are reviewing the abstract). We suggest leaving them in and making the decisions at the time(s) of detailed coding, as the coding itself will often prove if the paper is relevant or not. The screening results should be validated by another person. Often this can be done by simply taking a sample of the papers in each pool (i.e. from the 'included' lot and the 'excluded' lots) and requesting a review by someone who is not involved in classification stage.

3.5. Paper extraction summary

The main aim of this step is to create a log of how the papers were extracted to form the final pool that goes into the analysis phase. This should include the multiple rounds of searching and their results, both pre- and post-removal of duplications, any efforts of forward and backward searching, and the outcomes before and after the screening rounds (as described in 3.4 above). Ideally, this is captured in a simple visual form that is easy to communicate. Examples of such visualisations are decision trees, simple flowcharts, or tabulations. This kind of summary can also be presented as part of the main paper, of which we have seen some prior examples (e.g. Figure 1 of Thennakoon et al. (2018, p. 481). Such visualisations give an easy-to-digest overview on the paper selection efforts made by the authors.

3.6. Preparing for coding

This section bridges the paper-extraction phase with the coding phase. It is a critical intermediate step that forms the foundations on which the coding (in the next stages) is built upon. Some key decisions need to be made here – namely about the use of tools (is a tool being used at all? Which tool? How should it be set up?), and the overall coding plan.

3.6.1. Tool use

While literature reviews need not always be supported by tools, we *do* recommend considering the plethora of available qualitative data analysis tools (such as NVivo, Leximancer etc.) which have dedicated features to support literature reviews. They often are accompanied with free online learning material, which makes upskilling on the topic relatively easy. The selected tool's features and affordances need to align with the overall study goals. There are papers dedicated to discussing tool support in literature reviews, such as Bandara et al. (2015), which can be used here. Tools such as NVivo also offers additional functionality such as cloud collaboration that enables researchers to perform coding simultaneously.

For the protocol, it is important that you clearly state (i) if a tool is used, if so (ii) what tool, (iii) why it was selected, and also confirm (iv) the tool features that are likely to be used (see [Appendix A – 6.1](#)). Key considerations and essential steps to follow in the setup and eventual use of the tool, should also be captured in step (iv). Another important element is to confirm that the research team has access to the tool and also has the needed capabilities to use the suggested tool and its features. If not, mechanisms to address access or skills gaps need to be considered and executed.

In the absence of a tool, the researchers are still bound to use some kind of systematic approach to help manage the data extraction and analysis process. For example, one may set up Excel spreadsheets where columns represent the diverse themes and the rows show the corresponding papers (similar to the thematic analysis approach discussed by Webster and Watson (2002). We recommend that these procedures and the supporting templates (in this case the draft spreadsheet(s)) are recorded in the protocol, and are discussed and quality checked with multiple researchers.

3.6.2. Overall coding plans

Literature reviews will follow typical qualitative data analysis and coding approaches. Coding is the process by which we extrapolate the information that is relevant for our research – information is sorted into ‘codes’ from the pool of papers and synthesised to make meaning. ‘codes’, by definition, are ‘tags or labels for assigning units of meaning to the descriptive or inferential information compiled during a study’ (Miles & Huberman, 1994, p. 56). Codes can be developed from prior theory, or they can emerge from the data inductively.

Coding will occur in different steps that we refer to as ‘coding cycles’ (following Saldaña, 2015). Having a plan on what these cycles would be (accepting that they *will* evolve during the actual coding process) and capturing and communicating this through the literature review protocol is highly recommended (see [Appendix A – 6.2](#)). This involves identifying the key cycles of coding and briefly describing what is due to take place within each coding cycle. Saldana et al. (2011), in part two of their book, vividly discuss these multiple cycles of coding, and provide detailed techniques on how to progressively refine codes.

We recommend running a ‘pilot’ round of coding and exposing this to reflection and coder corroboration as each coding cycle is commenced. We also suggest determining who is involved in the activity and pre-plan the role(s) of diverse coders within the research team and how they will contribute. The contributions can be towards the actual coding, or towards quality assurance in the planned coder-corroboration sessions (which are recommended to occur at the end of each coding cycle).

The crafting of coding rules and codebooks is also essential in this stage. A researcher can begin constructing principles for text-to-code translation. This will keep the coding process uniform and orderly. The researcher can code precisely what is desired. The validity of the coding process is ensured when the researcher’s codes are consistent and coherent i.e. when they adhere to their coding rules. **Coding rules** are the guidelines which provide directions that the coders must follow during the act of coding. For example, a coding rule may capture how pieces of data are extracted from the original paper and assigned to specific codes, whether it is by fragments, sentences, or paragraphs (see [Appendix A – 6.3](#) and [Appendix B – Part B, Section 3](#) in Ancillary material). Coding rules can also detail minor complications, like if the same content can be coded under more than one code or provide information on how coding memos are expected to be maintained.

A code book by definition is ‘a set of codes, definitions and examples used as a guide to help analyse’ the data (DeCuir-Gunby et al., 2011, p. 137). Code books are developed through an iterative process and are bound to evolve as the coding progresses. Deductive (i.e. theory-driven) coding approaches are likely to have more detailed coding books upfront, that are based on the a priori theory, whereas inductive (i.e. data-driven) approaches form the codebooks as the analysis occurs. Regardless of the coding approach, it is highly recommended that these code books are derived, maintained, and reported on as the study findings are presented. DeCuir-Gunby et al. (2011) provide in-depth details about the derivation of codebooks. Though these guidelines are presented for other qualitative research methods such as interviews, they are applicable to literature reviews as well. While diverse details can be captured within a code book, the bare minimum should capture the details for

each code by using the following requirements: (i) a clear label, (ii) a detailed definition and description for the code which includes inclusion/exclusion criteria, and (iii) an example (see [Appendix A – 6.4](#)).

Assuring the reader of the quality of your coding approach is indeed an important aspect of this phase. The coding rules, code books, pilot sessions etc., and how these are designed, executed, and reported on, will play a key role in assuring coding quality. Intercooder reliability is another common method that qualitative researchers use, especially within content analysis efforts, to try to establish coding quality. When a literature review closely follows a content analysis paradigm (e.g. for a ‘meta-analysis’), then reporting of intercooder reliability is of value. Nili et al. (2020) provide comprehensive guidelines on selecting and using different intercooder reliability approaches which can be deployed to literature reviews as well. When a literature review applies a ‘grounded theory’ approach (Wolfswinkel et al., 2013), for example, to a ‘theoretical review’ or even to a more interpretative paradigm (in other types of literature reviews), then it is accepted that different coders’ will organise codes into different themes based on the coders’ diverse theoretical backgrounds. These can be followed by coder corroboration sessions, where group discussions of divergences and overlaps are discussed and ironed out (O’Connor & Joffe, 2020).

3.7. Preparing to report

The purpose of this part of the literature review protocol is to encourage the researchers to think beyond the analysis and form a vision on what the final results may look like. This fulfils multiple purposes. It pushes the researcher to think deeply about the findings in advance, including the knowledge contributions the findings are meant to deliver, the different fields of data and evidence that are needed to report the anticipated findings, and the optimal ways on which to present the findings – for example, how to best blend text, tables, and figures, while maintaining a balance between results, interpretations, and evidence. This forward thinking about the end results at the time of initial protocol design enables the researcher to better pre-plan towards achieving these target outcomes more efficiently. Miles and Huberman (1994) present a rich reference source to ideate diverse ways in how the final results can be presented. We recommend to the readers to refer to this resource and, together with the set goals of the literature review (as articulated in [Section 3.1](#)), ideate and document some ‘skeleton templates’ to draft how the results would be presented. This also provides mental comfort, especially for novice researchers, who can carry on with ease knowing what the end result will look like.

4. Discussion & Conclusion

This paper contributes to the existing discourse of literature review methodological guidelines by introducing the role of a literature review protocol. In any literature review effort, the benefits of using and reporting on a protocol are many, such as propagating transparency, quality assurance, and efficiency. In addition to outlining in detail the elements that should exist in a literature review protocol, which was

complemented by templates, examples and supporting references, we make a few further recommendations:

- **Deeply embed and integrate the protocol with the overall study design** of a literature review paper. In a literature review paper's methodology section, dedicate space to describe the key aspects covered in the protocol, how the protocol design took place, and how it contributed towards the steps within the methodology. As examples, see Li and Chan (2019), Syed et al., (2020) and Syed et al., (2018).
- Use the protocol not only for internal purposes within the research team, but also **share the protocol as part of your literature review paper**, as this will increase the reliability of the findings presented. In our roles as literature review paper reviewers and editors, we often notice how the protocol is sometimes shared only within the review process, where either its existence is merely mentioned in just one statement (e.g., Hill and de Zubielqui (2022), or the authors state that the protocol is 'available upon request' (e.g., Bharosa et al., 2009). On the contrary, we suggest that a well-compiled protocol is published *with the paper*. Space is often a constraint here, but this can be effectively overcome by integrating the protocol into additional supplementary material that can be accessed via external links. Furthermore, a well-presented protocol can be adapted beyond a single study. For example, Li and Chan (2019, p. 6) refer to and use Joshi et al.'s (2010) protocol, despite the disparate domains the two studies are from. Thus, sharing a protocol as part of a literature review paper can bring multiple benefits to the authors.
- Be open to the fact that **how you adopt and use the generic protocol template offered here will differ** from study to study based on targeted literature review type, overall study goals, available literature, and intended target audiences and outlets. All applications should carefully consider and integrate any contextual nuances during the adaption of the resources provided herein.
- **The protocol is a living document.** While it is recommended practice to develop the protocol as much as possible at the outset of the study, many elements within the protocol will evolve during the actioning of it. Accept this nature and be open to iterative changes. One can also use the **protocol as a project management tool** within a literature review initiative. For example, Syed et al., (2020) used the protocol for work allocation, maintaining coding consistency, enabling the credibility and transparency of the coding process, and to develop the team's coding skills (see [Appendix B Part B](#) for further details).

In conclusion, we hope this paper becomes a source of inspiration and point of reference for both novice and experienced literature review researchers to design, deploy, and communicate the literature review protocol, which we argue forms a critical part of any literature review effort. This paper will also benefit reviewers and editors, a sit can be used to check that the essential elements of a literature review process have been appropriately designed and communicated. Research method training educators may also find this paper valuable, as it forms a rich resource kit that can be used in trainings on how to conduct literature reviews.

Notes

1. ACPHIS PhD Medal is awarded annually for the **best Australian PhD thesis** in the discipline of Information Systems
2. *Inductive/deductive/hybrid*.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Appendix A:Sample Template of a Structured Literature Review (SLR) Protocol

Note: Each of these elements presented in the template below are described more vividly in the main paper (see [Section 3](#) above). You may refer to the detailed illustrative case example provided in [Appendix B](#) Section B to see how this generic protocol template and its elements can be adapted to the context of a specific study.

1. Goal (or purpose) of the literature review

[See [Section 1](#) of the illustrative case example in Ancillary material part B for an example of an adapted version]

<Write a few lines that describe what you would like to do/achieve from your literature review>

If you have research questions list them below:

- 1.
- 2.
- 3.
- ... ?

Alternatively, provide a few goal statements (~3) that summarise the key things you would like to achieve from your literature review efforts. Reflect on the mini summary you provided above to derive these.

- 1.
- 2.
- 3.
- ... ?

2. Key phenomena of interest

[In the illustrative case example (see Ancillary material part B), the phenomenon of interest was Robotic Process Automation, or RPA. Synonyms and related concepts included concepts such as Intelligent process automation, virtual workforce, software robots etc.]

Key Phenomenon/topics	Synonyms and related concepts
1.	
2	
...	

3. The search strategy

[See [Section 2](#) of the illustrative case example in Ancillary material part B for an example of an adapted version]

3.1 *Where do you search in?*

List and justify:

Which sources will you search in?	Rationale

Which fields will you include in your search?	Rationale

3.2 *Criteria for initial search*

Criteria at initial search	Justification

3.3 Search strings

Outlet	Search String(s) used	Outcomes derived	Comments

4. Paper screening

[See [Table 2](#) or [Section 2](#) of the illustrative case example in Ancillary material part B]

Overall stages of Screening:

<Describe them briefly as Round 1 to Round N, so the process is clear>

.....

.....

.....

.....
.....

Capture decision-making heuristics for each round:

Round 1 Screening:

.....
.....
.....
.....

Screening Criteria	Rationale?

<Note the results when executing the process>
<Repeat for all other rounds ... >

5. Paper Extraction Summary

[See Figure 3 of the illustrative case example in Ancillary material part B for an example of an adapted version]

<Visually capture the paper extraction key steps and outcomes here>

6. Preparing for coding

[See Sections 3 – 6 of the illustrative case example in Ancillary material part B for an example of an adapted version]

6.1 Tool use

- Will a tool be used to support the review process? Yes no
- If so what tool(s)?
 - Provide a rationale for their selection
- Confirmation that the full research team has access to the tool:
.....
- Confirmation that the full research team has the essential capabilities to use the tool:
.....
.....
- Tool features

Tool features planned to be used	Method of usage

- Tool set up
- <List the key steps on how the Tool will be set up for the project>

6.2 High level coding process/plan

Cycle of Coding	Coding approach? ²	Pilot coding plans	Related Coding rules	Who is involved & how?

6.3 Coding Rules

R1:
.....
.....
.....
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R2:
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.....
.....
.....
Rn:
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.....
.....
.....

6.4 Coding book(s)

Code	Definition (or description)	Example

6.5. Coder corroborations and/or intercoder reliability checks

Decide whether to have coder corroboration sessions, run intercoder reliability checks, or action both.

For coder corroboration sessions:

- Plan the corroboration sessions with a clear purpose and agenda.
- Make sure each coder comes prepared to present and justify their coding after having reviewed the other coders’ outcomes.
- Set up clear heuristics and procedures to confirm codes.

Intercoder reliability checks:

- Determine the intercoder reliability check to be used.
- Ensure that the coding scheme is clearly defined.

- Make sure that coders are trained (and on the same page).
 - Do a pilot test to assess this.
- Calculate intercoder reliability.
- Develop and deploy mechanisms to resolve discrepancies.
- Capture and report the process.

7. Reporting plans

[Have a look at the published final paper of Syed et al. (2020) and the protocol presented as the illustrative case example in Ancillary material part B to view the parallels.]

Over here, make a list of things that you anticipate to report as a result of the analysis, and also try to visualise how it will be presented. For example (i) the combination of text, tables, figures that cover (ii) final results, interpretations and evidence.

- *Draft a proposed structure and template of what your findings may look like.*
- *See if any new or previously unidentified aspects may be reported based on the findings you have.*
- *Cross-check that (a) the fields you want to report about and (b) the data that you have planned to capture in your codes and meta-data align. Confirm that you have what you need to report accordingly. Otherwise, adjust the earlier coding plans to fill the gaps.*

Appendix B: Ancillary Material

The content here is presented as Ancillary Material, available at this Google Drive link: [https://drive.google.com/drive/folders/1rmbR0iesDoYN3_37YspzrK5lZcAXC6OI]. It consists of two parts:

Part A lists and describes several seminal literature review guidelines that we recommend as useful resources to review prior to developing your own literature review guidelines.

Part B presents illustrative examples of the concepts discussed in this paper.